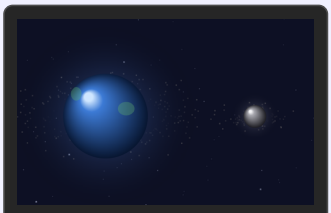
Lab 4: The Formation of the Moon

Four ideas, one Moon. You are the scientists.

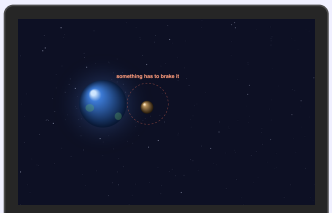
The Moon is odd: about a quarter of Earth's size, far bigger for its planet than almost any other moon. Test each idea against real data from Apollo rocks, lunar orbiters and meteorites, then make your case in a one-page lab report.

The four ideas



1 Sister Planet

The Moon formed next to Earth, at the same time, from the same part of the disk.



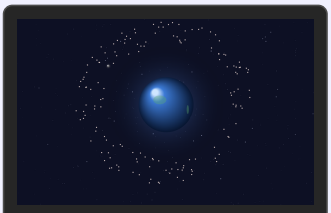
2 Captured Asteroid

The Moon formed somewhere else, and Earth's gravity captured it as it passed.



3 Fission Fragment

The young Earth spun so fast that a piece of its outer layers flew off.



4 Giant Impact

A Mars-sized planet, Theia, hit the young Earth. Rock from the outer layers splashed into a ring that clumped into the Moon.

PART 1: EXPLORE (the Lab 4 app, on the class site)

1. **Four Hypotheses.** Pick an idea and read what it predicts. Each card says *if this idea is right, we'd see…*. Open the data, then judge: **does the data match the prediction?** ✓ **matches** = supports the idea, × **doesn't match** = argues against it. Do all four ideas.
2. **Theia.** Launch at least three impacts, changing one thing at a time (angle, mass or speed). Follow the hints. Which hit makes a Moon like ours: 1.2% of Earth's mass, with very little iron? Record the % of the Moon made from Theia.
3. **My Verdict.** Pick the idea the evidence supports. Print your evidence board and notes.

PART 2: WRITE IT UP (one page, as a group)

1. **Introduction** (2–3 sentences): why the Moon is odd, and the four ideas *in your own words.*
2. **Procedure** (1 sentence): *"We used data obtained and validated by the scientific community, explored with the Lab 4 app."*
3. **Results:** your evidence board as a table (4 ideas × 6 kinds of evidence), plus one figure from the app, captioned, axes labeled. Cite it (astroclass, 2026).
4. **Conclusion:** (a) your claim; (b) the one or two pieces of evidence that convinced you; (c) for each other idea, the best evidence against it; (d) one thing still unexplained, and what data would settle it.

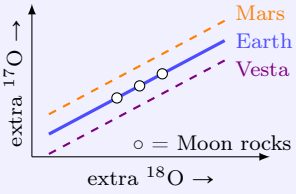
Words you'll see

Look for these in the app. Each one goes with a tab.

Fingerprints tab — oxygen isotopes



8 protons each, 8, 9 or 10 neutrons



Isotopes: the same element with different numbers of neutrons. Each part of the early solar system got its own mix.

Isotope line: every rock from one body falls on one line on this plot.

ppm: parts per million. $1\,‰ = 1000$ ppm.

Iron tab — what sinks, what stays



1. cold, all mixed

2. it **melts**: siderophiles sink, lithophiles stay

3. iron core, rocky mantle

Differentiation: a young planet melts; heavy iron sinks to the core, light rock stays on top.

Density: iron-rich planet $\approx 5\,\mathrm{g/cm^3}$; rock alone ≈ 3.

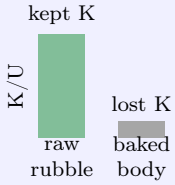
Chondrite: a meteorite that never melted: the raw rubble planets were built from.

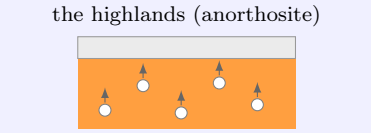
If a planet melts, its siderophiles sink to the core. The rock left on top is poor in iron and nickel, but keeps its lithophiles.

Siderophile "iron-loving": dissolves in melted iron and **sinks to the core**. *Fe, Ni, Au, Pt*

Lithophile "rock-loving": stays in the rock, **on top**. *Si, Mg, Al, K*

Baked Dry and Two Faces tabs — a very hot start

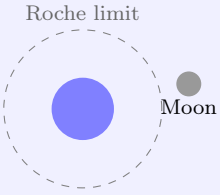


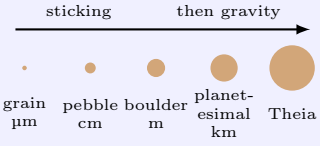


Volatile: boils away easily (water, potassium K). **Refractory:** almost never does (uranium U). Low **K/U** = baked dry.

Magma ocean: an outer layer hundreds of km deep, all melted at once.

Spin, Size and Theia tabs





Angular momentum: spin; it is conserved, $L = I\omega$.

Roche limit: too close to a planet, tides pull clumps apart.

Mass ratio: moon's mass / planet's mass. Ours: 1/81.

Planetesimal: a km-sized building block. **Theia** was a leftover planet.